HP-Azithromycin Powder for Solution for Infusion 500mg

Composition:

One vial contains 500 mg azithromycin (as dihydrate). Provides 100 mg/ml concentrate for reconstitution. The concentrate is further diluted to 1 mg/ml or 2 mg/ml.

Product Description:

White lyophilized powder supplied in 10 mL sterile vial.

After reconstitution/dilution: Colorless clear liquid.

Pharmacological properties

Pharmacodynamic properties

Azithromycin is the first of a subclass of macrolides antibiotics, known as azalides, and is chemically different from erythromycin. Chemically it is derived by insertion of a nitrogen atom into the lactone ring of erythromycin A. The mode of action of azithromycin is inhibition of protein synthesis in bacteria by binding to the 50s ribosomal subunit and preventing translocation of peptides.

Azithromycin demonstrates activity in vitro against a wide range of bacteria including:

<u>Gram-positive Aerobic Bacteria</u>

Staphylococcus aureus, Streptococcus pyogenes (group A beta-hemolytic streptococci), *Streptococcus pneumonia, alpha-hemolytic streptococci* (viridans group) and other streptococci, and *Corynebacterium diphtheria*.

Azithromycin demonstrates cross-resistance with erythromycin-resistant Gram-positive strains, including *Streptococcus faecalis* (enterococcus) and most strains of methicillin-resistant staphylococci.

<u>Gram-negative Aerobic Bacteria</u>

Haemophilus influenzae, Haemophilus parainfluenzae, Moraxella catarrhalis, Acinetobacter species, *Yersinia* species, *Legionella pneumophila, Bordetella pertussis, Bordetella parapertussis, Shigella species, Pasteurella species, Vibrio cholerae* and *parahaemolyticus, Plesiomonas shigelloides*. Activities against *Escherichia coli, Salmonella enteritidis, Salmonella typhi*, *Enterobacter* species, *Aeromonas hydrophila* and *Klebsiella* species are variable and susceptibility tests should be performed. *Proteus* species, *Serratia* species, *Morganella* species, and *Pseudomonas aeruginosa* are usually resistant.

<u>Anaerobic Bacteria</u>

Bacteroides fragilis and *Bacteroides* species, *Clostridium perfringens*, *Peptococcus* species and *Peptostreptococcus* species, *Fusobacterium necrophorum* and *Propionibacterium acnes*.

<u>Organisms of Sexually Transmitted Diseases</u>

Azithromycin is active against Chlamydia *trachomatis* and also shows good activity against *Treponema pallidum, Neisseria gonorrhoeae,* and *Haemophilus ducreyi*.

<u>Other Organisms</u>

Borrelia burgdorferi (Lyme disease agent), *Chlamydia pneumonia*, *Mycoplasma pneumoniae*, *Mycoplasma hominis, Ureaplasma urealyticum, Campylobacter* species and *Listeria monocytogenes*.

<u>Opportunistic Pathogens Associated with HIV Infections</u>

Mycobacterium avium-intracellulare complex, Pneumocystis carinii and *Toxoplasma gondii*.

Commonly susceptible species:

<u>Aerobic Gram-positive bacteria</u>

Staphylococcus aureus, *Streptococcus agalactiae*, Streptococci (Groups C, F, and G) and Viridans group streptococci.

<u>Aerobic Gram-negative bacteria</u>

Bordetella pertussis, *Haemophilus ducreyi*, *Haemophilus influenzae*, *Haemophilus parainfluenzae*, *Legionella pneumophila*, *Moraxella catarrhalis* and *Neisseria gonorrhoeae*.

Other

Chlamydia pneumoniae, *Chlamydia trachomatis*, *Mycoplasma pneumonia* and *Ureaplasma urealyticum*.

Species for which acquired resistance:

Aerobic Gram-positive bacteria

Streptococcus pneumoniae, Streptococcus pyogenes

Note: Azithromycin demonstrates cross-resistance with erythromycin-resistant gram-positive strains. Inherently resistant organisms: Enterobacteriaceae, *Pseudomonas.*

Pharmacokinetic properties

Absorption

Following oral administration in humans, azithromycin is widely distributed throughout the human body; bioavailability is approximately 37%. The time taken to peak plasma levels is 2-3 hours.

Distribution

High azithromycin concentrations have been observed inphagocytes. In experimental models, higher concentrations of azithromycin are released during active phagocytosis than from non- stimulated phagocytes.This results in high concentrations of azithromycin being delivered to the site of infection.

Pharmacokinetic studies in humans have shown markedly higher azithromycin levels in tissue than in plasma (up to 50 times the maximum observed concentration in plasma) indicating that the drug is heavily tissue bound. Concentrations in target tissues, such as lung, tonsil and prostate exceed the MIC90 for likely pathogens after a single dose of 500 mg.

Elimination

Plasma terminal elimination half-life closely reflects the tissue depletion half-life of 2 to 4 days. Approximately 12% of an intravenously administered dose is excreted in the urine over 3 days as the parent drug, the majority in the first 24 hours. Biliary excretion of azithromycin is a major route of elimination for unchanged drug following oral administration. Very high concentrations of unchanged drug have been found in human bile, together with 10 metabolites, formed by N- and O-demethylation, by hydroxylation of the desosamine and aglycone rings, and by cleavage of the cladinose conjugate. Comparison of HPLC and microbiological assays in tissues suggests that metabolites play no part in the microbiological activity of azithromycin.

Pharmacokinetics in Special Patient Groups

Elderly

In elderly volunteers (> 65 years), slightly higher AUC values were seen after a 5-day regimen than in young volunteers (< 40 years), but these are not considered clinically significant, and hence no dose adjustment is recommended.

Renal Impairment

The pharmacokinetics of azithromycin in subjects with mild to moderate renal impairment (GFR 10 - 80 ml/min) were not affected following a single one gram dose of immediate release azithromycin. Statistically significant differences in AUC 0-120 (8.8 mcg·h/ml vs. 11.7 mcg·h/ml), C_{max} (1.0 mcg/ml vs. 1.6 mcg/ml) and CLr (2.3 ml/min/kg vs. 0.2 ml/min/kg) were observed between the group with severe renal impairment (GFR < 10 ml/min) and the group with normal renal function.

Hepatic Impairment

In patients with mild to moderate hepatic impairment, there is no evidence of a marked change in serum pharmacokinetics of azithromycin compared to those with normal hepatic function. In these patients urinary clearance of azithromycin appears to increase, perhaps to compensate for reduced hepatic clearance.

Indication

HP-Azithromycin Powder for Solution for Injection 500mg is indicated for the treatment of community acquired pneumonia (CAP) caused by susceptible organisms, including *Legionella pneumophila*, in patients who require initial intravenous therapy.

HP-Azithromycin Powder for Solution for Injection 500mg is indicated for the treatment of pelvic inflammatory disease (PID) caused by susceptible organisms (*Chlamydia trachomatis, Neisseria gonorrhoea, Mycoplasma hominis*), in patients who require initial intravenous therapy.

Recommended dosage

Adults: For the treatment of adult patients with CAP due to the indicated organisms, the recommended dose of intravenous azithromycin is 500 mg as a single daily dose by the IV route for at least 2 days. IV therapy should be followed by oral azithromycin as a single daily dose of 500 mg to complete a 7 to 10 day course of therapy. The timing of the conversion to oral therapy should be done at the discretion of the physician and in accordance with clinical response.

For the treatment of adult patients with PID due to the indicated organisms, the recommended dose of intravenous azithromycin is 500 mg as a single daily dose by the IV route for one or two days. IV therapy should be followed by azithromycin by the oral route at a single daily dose of 250 mg to complete a 7-day course of therapy. The timing of the conversion to oral therapy should be done at the discretion of the physician and in accordance with clinical response. If anaerobic microorganisms are suspected of contributing to the infection, an antimicrobial anaerobic agent may be administered in combination with azithromycin.

Children: The safety and efficacy of intravenous azithromycin for the treatment of infections in children has not been established.

Elderly: The same dosage as in adult patients is used in the elderly. Elderly patients may be more susceptible to the development of torsades de pointes arrhythmia than younger patients.

Patients with Renal Impairment:

No dosage adjustment is necessary in patients with mild to moderate renal impairment (GFR 10 – 80 ml/min). Caution should be exercised when azithromycin is administered to patients with severe renal impairment (GFR < 10 ml/min).

Patients with Hepatic Impairment:

The same dosage as in patients with normal hepatic function may be used in patients with mild to moderate hepatic impairment.

Route of Administration: INTRAVENOUS (IV)

<u>Method of administration</u>

Administered by intravenous infusion over 3 hours with a concentration of 1 mg/ml or over 1 hour with a concentration of 2 mg/ml. Higher concentrations should be avoided, as all subjects that received infusion concentrations above 2 mg/ml experienced local reactions at the infusion site. The infusion time for azithromycin should not be shorter than 60 minutes.

HP-Azithromycin Powder for Solution for Injection 500mgshould not be given as a bolus or intramuscular injection.

<u>For instructions on reconstitution of the medicinal product before administration:</u>

HP-Azithromycin Powder for Solution for Injection 500mgis supplied in 10 ml single-use vials with 500 mg substance.

Step 1

Prepare stock solution by adding 4.8 ml of sterile water for injections into the vial of HP-Azithromycin Powder for Solution for Injection 500mg. Shake the bottle until all powder is dissolved. 1 ml of reconstituted solution contains 100 mg azithromycin.

Step 2

Dilute the resulting 5 ml premix further with a compatible infusion to obtain the final infusion solution of azithromycin at a concentration of 1 mg/ml or 2 mg/ml (see Table below).

Preparation of final infusion solution

Concentration final infusion solution (mg/ml)	Quantity of Diluent
1 mg/ml	500 ml
2 mg/ml	250 ml

Infusion concentrate may be diluted with:

0.9% sodium chloride injection

0.45% sodium chloride injection

5% Dextrose in Water for Injection

Lactated Ringer's Injection

5% Dextrose in 0.45% sodium chloride injection

5% Dextrose in 0.3% sodium chloride injection

5% Dextrose in 0.45% sodium chloride with 20mEq potassium chloride

5% Dextrose in Lactated Ringer's Injection

Prior to administration, the vial should be inspected visually for particulate matter. If the solution contains particles, it should be discarded.

Contraindication

Hypersensitivity to azithromycin, erythromycin or any other macrolide- or ketolide antibiotics or to any of the excipients.

Warning and precaution

Hypersensitivity

As with erythromycin and other macrolides, rare serious allergic reactions, including angioedema and anaphylaxis (rarely fatal), dermatologic reactions including Stevens - Johnson syndrome (SJS), Toxic Epidermal Necrolysis (TEN) (rarely fatal), and Drug Reaction with Eosinophilia and Systemic Symptoms (DRESS) have been reported. Some of these reactions with azithromycin have resulted in recurrent symptoms and required a longer period of observation and treatment. If an allergic reaction occurs, the drug should be discontinued and appropriate therapy should be instituted. Physicians should be aware that reappearance of the allergic symptoms may occur when symptomatic therapy is discontinued.

Prolongation of the QT interval

Prolonged cardiac repolarization and QT interval, imparting a risk of developing cardiac arrhythmia and torsades de pointes, have been seen in treatment with macrolides, including azithromycin. Prescribers should consider the risk of QT prolongation, which can be fatal, when weighing the risks and benefits of azithromycin for at-risk groups including:

• Patients with congenital or documented QT prolongation

• Patients currently receiving treatment with other active substances known to prolong QT interval, such as antiarrhythmics of Classes IA and III, antipsychotic agents, antidepressants, and fluoroquinolones

• Patients with electrolyte disturbance, particularly in cases of hypokalemia and hypomagnesemia

• Patients with clinically relevant bradycardia, cardiac arrhythmia or cardiac insufficiency

• Elderly patients: elderly patients may be more susceptible to drug associated effects on the QT interval. Since liver is the principal route of elimination for azithromycin, the use of azithromycin should be undertaken with caution in patients with significant hepatic disease.

In patients receiving ergot derivatives, ergotism has been precipitated by coadministration of some macrolide antibiotics. There are no data concerning the possibility of an interaction between ergot and azithromycin. However, because of the theoretical possibility of ergotism, azithromycin and ergot derivatives should not be coadministered.

As with any antibiotic preparation, observation for signs of super infection with non-susceptible organisms, including fungi is recommended.

Clostridium difficile associated diarrhea (CDAD) has been reported with use of nearly all antibacterial agents, including azithromycin, and may range in severity from mild diarrhea to fatal colitis. Treatment with antibacterial agents alters the normal flora of the colon leading to overgrowth of *C difficile*.

C. difficile produces toxins A and B which contribute to the development of CDAD. Hypertoxin producing strains of *C. difficile* cause increased morbidity and mortality, as these infections can be refractory to antimicrobial therapy and may require colectomy. CDAD must be considered in all patients who present with diarrhea following antibiotic use. Careful medical history is necessary since CDAD has been reported to occur over two months after the administration of antibacterial agents.

In patients with severe renal impairment (GFR<10 ml/min) a 33% increase in systemic exposure to azithromycin was observed.

Infantile hypertrophic pyloric stenosis (IHPS) has been reported following the use of azithromycin in infants (treatment up to 42 days of life). Parents and caregivers should be informed to contact their physician if vomiting and/or irritability with feeding occurs.

Interactions with Other Medicaments

Antacids

No effect on overall bioavailability was seen although peak serum concentrations were reduced by approximately 25%. In patients receiving both oral azithromycin and antacids, the drugs should not be taken simultaneously.

Cetirizine

Coadministration of a 5-day regimen of azithromycin with cetirizine 20mg at steady-state resulted in no pharmacokinetic interaction and no significant changes in the QT interval.

Didanosine (Dideoxyinosine)

Coadministration of 1200 mg/day azithromycin with 400 mg/day didanosine in HIV-positive subjects did not appear to affect the steady-state pharmacokinetics of didanosine as compared with placebo.

Digoxin and colchicines

Concomitant administration of macrolide antibiotics, including azithromycin, with P-glycoprotein substrates such as digoxin and colchicine, has been reported to result in increased serum levels of the P-glycoprotein substrate. Therefore, if azithromycin and P-glycoprotein substrates such as digoxin are administered concomitantly, the possibility of elevated serum digoxin concentrations should be considered. Clinical monitoring, and possibly serum digoxin levels, during treatment with azithromycin and after its discontinuation are necessary.

Zidovudine

Single 1000 mg doses and multiple 1200 mg or 600 mg doses of azithromycin had little effect on the plasma pharmacokinetics or urinary excretion of zidovudine or its glucuronide metabolite. However, administration of azithromycin increased the concentrations of phosphorylated zidovudine, the clinically active metabolite, in peripheral blood mononuclear cells. The clinical significance of this finding is unclear, but it may be of benefit to patients.

Azithromycin does not interact significantly with the hepatic cytochrome P450 system. It is not believed to undergo the pharmacokinetic drug interactions as seen with erythromycin and other macrolides. Hepatic cytochrome P450 induction or inactivation via cytochrome-metabolite complex does not occur with azithromycin.

Ergot

Due to the theoretical possibility of ergotism, the concurrent use of azithromycin with ergot derivatives is not recommended.

Atorvastatin

Coadministration of atorvastatin (10 mg daily) and azithromycin (500mg daily) did not alter the plasma concentrations of atorvastatin.

Carbamazepine

No significant effect was observed on the plasma levels of carbamazepine or its active metabolite in patients receiving concomitant azithromycin.

Cimetidine

No alteration of azithromycin pharmacokinetics was observed when given a single dose of cimetidine 2 hours before azithromycin.

Coumarin-Type Oral Anticoagulants

There have been reports of potentiated anticoagulation subsequent to coadministration of azithromycin and coumarin-type oral anticoagulants. Although a causal relationship has not been established, consideration should be given to the frequency of monitoring prothrombin time when azithromycin is used in patients receiving coumarin-type oral anticoagulants.

Cyclosporin

Caution should be exercised before considering concurrent administration of cyclosporin and azithromycin. If coadministration of these drugs is necessary, cyclosporin levels should be monitored and the dose adjusted accordingly.

Efavirenz

Coadministration of a 600 mg single dose of azithromycin and 400mg efavirenz daily for 7 days did not result in any clinically significant pharmacokinetic interactions.

Fluconazole

Coadministration of a single dose of 1200 mg azithromycin did not alter the pharmacokinetics of a single dose of 800 mg fluconazole. Total exposuremand half-life of azithromycin were unchanged by the coadministration of fluconazole, however a clinically insignificant decrease in C_{max} (18%) of azithromycin was observed.

Indinavir

Coadministration of a single dose of 1200 mg azithromycin had no statistically significant effect on the pharmacokinetics of indinavir administered as 800 mg three times daily for 5 days.

Methylprednisolone

Azithromycin had no significant effect on the pharmacokinetics of methylprednisolone.

Midazolam

Coadministration of azithromycin 500 mg/day for 3 days did not cause clinically significant changes in the pharmacokinetics and pharmacodynamics of a single 15mg dose of midazolam.

Nelfinavir

Coadministration of azithromycin (1200mg) and nelfinavir at steady state (750mg three times daily) resulted in increased azithromycin concentrations. No clinically significant adverse effects were observed and no dose adjustment is required.

Rifabutin

Coadministration of azithromycin and rifabutin did not affect the serum concentrations of either drug. Neutropenia was observed in subjects receiving concomitant treatment of azithromycin and rifabutin. Although neutropenia has been associated with the use of rifabutin, a causal relationship to combination with azithromycin has not been established.

Sildenafil

There was no evidence of an effect of azithromycin (500mg daily for 3days) on the AUC and C_{max}, of sildenafil or its major circulating metabolite.

Terfenadine

There have been rare cases reported where the possibility of such an interaction could not be entirely excluded; however, there was no specific evidence that such an interaction had occurred.

Theophylline

There is no evidence of a clinically significant pharmacokinetic interaction when azithromycin and theophylline are co-administered.

Triazolam

Coadministration of azithromycin with 0.125 mg triazolam had no significant effect on any of the pharmacokinetic variables for triazolam compared to triazolam and placebo.

Trimethoprim/sulfamethoxazole

Coadministration of trimethoprim/sulfamethoxazole DS (160mg/800mg) for 7 days with azithromycin 1200mg on Day 7 had no significant effect on peak concentrations, total exposure or urinary excretion of either trimethoprim or sulfamethoxazole. Azithromycin serum concentrations were similar to those seen in other studies.

Pregnancy and lactation

Pregnancy

There are no adequate data on the use of azithromycin in pregnant women. Safety of azithromycin has not been confirmed with respect to the active substance during pregnancy. Azithromycin should be given during pregnancy only if the benefit outweighs the risk.

Breast-feeding

Excretion of azithromycin in breast milk has been reported. Since it is not known whether azithromycin may have adverse effects on the nursing infant, breast-feeding should be discontinued during treatment with azithromycin. Among other effects, there is the risk of diarrhoea, fungus infection of the mucous membranes as well as sensitization of the infant.

Effects on ability to drive and use machines

There are no indications that azithromycin has produced any effect on the ability to drive or use machines.

Side effects

Azithromycin is well tolerated with a low incidence of side effects.

Blood and Lymphatic System Disorders

A causal relationship to azithromycin has not been established.

Ear and Labyrinth Disorders

Hearing impairment (including hearing loss, deafness and/or tinnitus) has been reported in some patients receiving azithromycin. Many of these have been associated with prolonged use of high doses in investigational studies. In those cases where follow-up information was available the majority of these events were reversible.

Gastrointestinal Disorders

Nausea, vomiting, diarrhea, loose stools, abdominal discomfort (pain/cramps), and flatulence.

Hepatobiliary Disorders

Abnormal liver function.

Skin and Subcutaneous Tissue Disorders

Skin and Subcutaneous Tissue Disorders:
Frequency not known: severe cutaneous adverse reactions (SCARs) including Stevens-Johnson syndrome (SJS), toxic epidermal necrolysis (TEN), drug reaction with eosinophilia and systemic symptoms (DRESS) & acute generalised exanthematous pustulosis (AGEP).

General Disorders and Administration Site Conditions

Local pain and inflammation at the site of infusion.

In post-marketing experience, the following additional undesirable effects have been reported:

Infections and Infestations

Moniliasis and vaginitis.

Blood and Lymphatic System Disorders

Thrombocytopenia.

Immune System Disorders

Anaphylaxis (rarely fatal)

Metabolism and Nutrition Disorders

Anorexia.

Psychiatric Disorders

Aggressive reaction, nervousness, agitation, and anxiety.

Nervous System Disorders

Dizziness, convulsions (as seen with other macrolides), headache, hyperactivity, hypoesthesia, paresthesia, somnolence, and syncope. There have been rare reports of taste/smell perversion and/or loss. However, a causal relationship has not been established.

Ear and Labyrinth Disorders

Vertigo.

Cardiac Disorders

Palpitations and arrhythmias including ventricular tachycardia (as seen with other macrolides) have been reported. There have been rare reports of QT prolongation and torsades de pointes (see Special warnings and precautions for use)

Vascular Disorders

Hypotension

Gastrointestinal Disorders

Vomiting/diarrhea (rarely resulting in dehydration), dyspepsia, constipation, pseudomembranous colitis, pancreatitis, and rare reports of tongue discoloration.

Hepatobiliary Disorders

Hepatitis and cholestatic jaundice have been reported, as well as rare cases of hepatic necrosis and hepatic failure, which have rarely resulted in death. However, a causal relationship has not been established.

Skin and Subcutaneous Tissue Disorders

Allergic reactions including pruritus, rash, photosensitivity, edema, urticaria, and angioedema. Rarely, serious cutaneous adverse reactions including erythema multiforme, SJS, TEN and DRESS have been reported.

Musculoskeletal and Connective Tissue Disorders

Arthralgia.

Renal and Urinary Disorders

Interstitial nephritis and acute renal failure.

General Disorders and Administration Site Conditions

Asthenia has been reported, although a causal relationship has not been established; fatigue, and malaise.

Symptoms and Treatment of Overdose

Adverse events experienced in higher than recommended doses were similar to those seen at normal doses. In the event of overdosage, general symptomatic and supportive measures are indicated as required.

Storage condition

Before reconstitution/dilution: Store below 30°C

After reconstitution/dilution:

Chemical and physical stability of the reconstituted concentrate has been demonstrated for 24 hours at 25°C. After reconstitution and dilution, the mixed solution is chemically and physically stable for 24 hours at or below 25°C or 7 days in the refrigerator (5°C). From a microbiological point of view, however, it is recommended that the product should be used directly.

If not used immediately, storage times and conditions are the responsibility of the user. It is normally no longer than 24 hours at 2-8°C, unless dilution has taken place in controlled and validated aseptic conditions.

Dosage and packaging:

500mg per vial × 10 vials

Manufacturer:

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